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Promoting well-being and preventing burnout in teacher education: a pilot study of a mindfulness-based programme for pre-service teachers in Hong Kong

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The stress that negatively affects teachers has been found to influence the turnover rate in the teaching profession. Recent research has shown that mindfulness-based programmes effectively promote well-being while addressing psychological distress. In this study, the authors investigated the effects of a six-week mindfulness-based programme on pre-service teachers in Hong Kong (N = 70). The authors hypothesised that a six-week mindfulness-based programme would increase mindfulness and well-being while reducing stress and symptoms of depression. Both intervention and control groups were assessed at baseline and post-intervention. A univariate analysis revealed a significant increase in mindfulness and well-being. A regression analysis indicated that mindfulness was a significant predictor of well-being, stress, anxiety and depressive symptoms. Qualitative data reflected that the mindfulness programme was beneficial to and feasible for pre-service teachers. Such a programme could be implemented in teaching education to promote well-being and stress management while preventing burnout among teaching professionals.

Keywords: mindfulness; well-being; stress; anxiety; depressive symptom

Introduction

Teaching has been identified as a high-stress profession in recent decades (Cacha 1981; Dunham 1992; Lambert and McCarthy 2006; Leung and Lee 2006). Adequate stress can generate proper learning and work motivation; however, overwhelming long-term stress can lead to negative influences on physical and mental health (Goldston et al. 1996; Muris et al. 2004). Research has suggested that stress is associated with various physical problems, such as high blood pressure, fatigue, headaches and insomnia, and mental problems, such as emotional exhaustion, depression, anxiety and outbursts of anger (Hauber et al. 1998; Kashani et al. 1995). Studies have shown that stressed-out teachers easily develop negative attitudes towards their colleagues and students, making them generally less capable of dealing with burnout and putting them at risk of quitting the profession. Schools, in turn, face the costs of replacing burned-out teachers with substitutes (Black 2004; Bowers 2004; Friedman 2000; Howard and Johnson 2004).

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The issue of teacher burnout is a cause for deep concern for school practitioners and policy makers in the West (Black 2004; Bowers 2004; Howard and Johnson 2004) and the East (Chan and Hui 1995; Lau 2002; Leung and Lee 2006; Tang et al. 2001). It is particularly alarming to see that increasing numbers of teachers have suffered from burnout in recent decades (Borg 1990; Vandengerghe and Huberman 1999). This is reflected in a study on teachers in Hong Kong schools, in which 40% of the sample were prepared to leave the profession and 50% were prepared to pursue new teaching opportunities (Wong 1989). These phenomena have been widely regarded by the public as directly associated with the introduction of a series of educational reforms that began in 2000 (Hong Kong Education Commission 2000). Since then, teachers' stress has reportedly intensified as they face increasingly stressful school environments. They must field constant requests from the school and the government to implement the new policy of Learning to Learn, take an active role in the new mechanism of internal and external evaluation of school performance, prepare for the new public examination system of the Hong Kong Diploma of Secondary Education and help shift school structures to accommodate the New Senior Secondary Education. Both professional and public awareness of teachers' stress has been raised by the newspaper headlines about incidents of young and experienced teachers committing suicide that have appeared throughout the last decade. The press has calculated that 11 teachers committed suicide at home or at school from 2006 to 2010 (*Oriental Daily* 2010). One study indicated that secondary school and language teachers suffered higher professional distress than five years ago when the accountability system had just been established (Hong Kong Primary Education Research Association and Education Convergence 2006). Another study indicated that teachers with less experience or lower salary were more likely to suffer from emotional exhaustion (Tang et al. 2001). New teachers in particular may be at high risk of suffering exhaustion and burnout as they tend to impose high standards and expectations on themselves. Many of them quit the teaching profession owing to physical and mental problems (Friedman 2000).

Various complex sources contribute to stress and burnout in the teaching profession, such as poor classroom management, a demanding curriculum, new policies, overwhelming paperwork, the principal's leadership style, lack of support from colleagues and personal health problems (Black 2004; Cacha 1981). Specifically, depersonalisation, personal accomplishment and emotional exhaustion were identified as major factors in burnout (Chan and Hui 1995; Maslach, Jackson, and Leiter 1996) and turnover intention (Lee and Asforth 1990). Relevant local studies have identified emotional exhaustion as the predominant predictor of turnover intention among Hong Kong teachers (Leung and Lee 2006; Liu and Wang 2004). It is partially caused by the cultural stereotypes of hard work, obedience and endurance which are strongly emphasised and commonly shared by people in Chinese society. The importance of developing a feasible intervention programme capable of reducing teachers' stress and emotional exhaustion has been emphasised in many studies (Chan and Hui 1995; Leung and Lee 2006; Mo 1991).

Studies have also found that positive emotions and health are associated with well-being (Huppert and Whittington 2003), which is defined as a 'combination of feeling good and functioning effectively' (Huppert 2009). Feeling good refers to positive experiences of happiness, emotions and affection, whereas functioning effectively relates to having a sense of purpose, value and autonomy (Huppert 2009; Huppert and Johnson 2010). In the field of mental health, the concept and approach of

enhancing well-being has recently transitioned from disorder prevention to mental health promotion (Huppert 2009). The same is true for the orientation of development in research and policy in this area (WHO 2001). Therefore, as an effective way of preventing burnout, teachers should learn how to enhance their well-being during the time when they receive their teacher training in universities or educational institutes. In short, encouraging pre-service teachers to join a programme that enhances well-being and stress reduction would be an effective strategy for preventing burnout.

Mindfulness-based programmes

In recent decades, mindfulness-based programmes have been recognised as effective interventions promoting well-being and reducing stress in both clinical and nonclinical populations. Studies have provided evidence supporting the effectiveness of mindfulness-based stress reduction (MBSR) in promoting health and well-being (Baer 2003; Brown and Ryan 2003; Williams et al. 2001), improving stress and anxiety (Kabat-Zinn 1990; Kabat-Zinn et al. 1992) and addressing emotional disorders (Arch and Craske 2006), attention-deficit problems (Brown and Ryan 2003), heart disease (Tacon et al. 2003), influenza antibodies (Davidson et al. 2003) and insomnia (Yook et al. 2008). Mindfulness-based cognitive therapy (MBCT) has been developed for the prevention of depressive relapses (Ma and Teasdale 2004; Segal, Williams, and Teasdale 2002; Teasdale et al. 2002; Williams 2008) and suicidal behaviour (Hargus et al. 2010; Lau, Segal, and Williams 2004). Some psychotherapies have also incorporated mindfulness for enhancing physical and mental well-being (Germer, Siegel, and Fulton 2005).

Mindfulness programmes are designed to enhance non-reactive awareness and openhearted acceptance (Kabat-Zinn 2003) by cultivating moment-to-moment attention and awareness of the existence of mind and body through feelings, sensations and thoughts (Bishop et al. 2004; Germer, Siegel, and Fulton 2005; Kabat-Zinn 1990). The effectiveness of mindfulness programmes has been reported in recent studies illustrating that primary school teachers who joined the MBSR programme reported significantly improved well-being and reduced stress and anxiety (Gold et al. 2010). However, thus far there have not been any studies on the effects of mindfulness programmes on the well-being of pre-service teachers in Hong Kong. Filling that gap was the aim of the pilot study reported here.

There were two goals in conducting a controlled trial for the mindfulness programme in the pilot study. The first was to evaluate such a programme's ability to improve the well-being and manage the stress, anxiety and depression of pre-service teachers in Hong Kong. The second was to evaluate the feasibility of offering the programme in a local setting. For the first goal, we hypothesised that the programme would result in an improvement in well-being and a reduction in stress, anxiety and depressive symptoms. For the second goal, we evaluated the feasibility and acceptability of the programme mainly from the qualitative feedback of the participants.

Methods

Participants and procedure

The participants were recruited from the Hong Kong Institute of Education, the largest teachers' training tertiary institute in Hong Kong. Having sought the approval

to conduct this study from the Ethical Review Committee of the Institute, publicity for the 'Mindfulness Education Programme' began in the second week of the first semester with posters placed throughout the campus and messages posted on the Institute's intranet. An oral presentation about the programme (5 to 10 minutes in length) was given by the project assistant to over 300 students from 20 classes. Participants who were interested in joining the project were asked to provide their names and contact information. One hundred and thirty-five students between the ages of 19 and 28 indicated preliminary interest in being part of a research project. The majority were enrolled in the Bachelor of Education Programme, with specialisations in English and Chinese language. The rest were evenly scattered over other specialisms such as mathematics, social studies, visual arts and music education. Those who expressed interest were then informed that they were free to join the study and could withdraw from it at any time. They were asked to prioritise their available timeslots according to their study schedules.

Students who enrolled in the mindfulness training group attended a six-week programme (15 hours) and a one-day retreat (7 hours). A 2.5-hour session per week would be offered in the Quiet Room on the university campus. This room was specially designed and renovated for meditation and prayer purposes. Owing to its limited size, only a maximum number of 26 participants could join each class. As 78 students indicated that they were available to join the six-week programme, they were divided into three groups. Accordingly, three classes delivered by the same instructor were offered for each group on three different days each week. Apart from the formal training sessions (a total of 15 hours for each group), all of the participants were invited to join a day-long retreat (7 hours for each group) scheduled on a Saturday in the fourth week that would include tea meditation, a mindful lunch and an outdoor mindful walking experience. Throughout the programme, the students were encouraged to do daily home practice on a voluntary basis (15 minutes/day).

After the first week, nine students dropped out of the classes owing to scheduling conflicts. The data gathered from 27 of the students were not acceptable, as they attended the programme for less than 12.5 hours. Forty-two students finished the six-week programme (15 hours), but seven did not attend the whole-day retreat (7 hours) and thus were not included in the final data set. Among the 57 students who showed interest in joining the control group, 22 were excluded from the final data because they did not complete the post-intervention questionnaire. Those who did complete the questionnaire were offered a book token.

Mindfulness programme

The participants undertook a six-week course delivered by the second author, an experienced secondary school teacher and a mindfulness instructor with more than 10 years of MBSR and insight meditation training experience. The six-week mindfulness programme, which was modified and adapted from the MBSR programme (Kabat-Zinn 1982, 1990), was presented to students as a short course to enhance well-being and concentration and reduce stress by cultivating moment-to-moment awareness. The programme was delivered using a secular approach that did not include any religious terminology. There were four major activities: (1) gentle stretching exercises that enhance the cultivation of mindfulness through an awareness of body movement and sensation; (2) daily mindfulness practice activities, including sitting, standing, walking and eating, all of which involve an awareness of

body sensations, thoughts and emotions; (3) a body scan consisting of a guided movement of attention throughout the body from the head to the toes while sitting or lying down; and (4) loving-kindness practice, which involves sending well wishes and blessings to oneself and all other people in the world.

The aims of the three major activities were to cultivate moment-to-moment awareness. For example, in a 15-minute walking practice, the participants were asked to be aware of the sensation as their feet touched the ground and the moving posture of each step. Owing to habitual mental patterns, most of the participants were distracted by some thoughts or emotions in the beginning of the walking practice. The instructor would remind them to focus on their feet. Most of the participants were able to enhance their calmness and concentration after 15 minutes of walking practice. The aim of the loving-kindness practice was to boost positive outlooks by cultivating friendship towards oneself and others. With the guidance of the instructor, participants would contemplate a few sentences in a sitting posture, i.e. 'May I be well and happy; may I be free from hatred; may I release my negative thoughts and emotions; may others be well and happy.' This practice was designed to increase self-acceptance, including past failures and inadequacies – a helpful tool for promoting well-being and the release of negativity brought on by depression and anxiety.

Regarding the design of our data collection, quantitatively the pre- and the post-intervention questionnaire surveys were conducted for the intervention and the control groups. Qualitative data about the student participants' experience of the mindfulness programme were also collected through focus group interviews. The interview method adopted for this study was proposed by Holstein and Gubrium (1995), in which the participants play equally significant roles in constructing a conversation. As a researcher, the interviewer felt obliged to show them respect and to care for their feelings and provide them with substantial freedom to talk about their experiences, clarifying that although the study was not intended to be secret or covert, the real names of the participants would be kept anonymous. In practice, the participants from each class were divided into five sub-groups and invited to share their experiences within their sub-groups for 10 minutes, followed by a whole-class sharing lasting five minutes. One of the groups would be randomly chosen and the conversations of its members would be recorded in the form of focus group interviews facilitated by a research assistant. The mindfulness instructor and the research assistant kept field notes of what the participants said when they shared in the whole-class and sub-group settings.

Furthermore, all of the participants were invited to write down their feedback on a written form that posed four questions: (1) 'Which activity made the strongest impression?'; (2) 'What kinds of feelings did you experience?'; (3) 'What kinds of thoughts did you observe?'; and (4) 'Other observations'. Furthermore, a Home Practice Record Form was also distributed to every participant to encourage them to do the suggested daily home practice.

For the analysis of qualitative data collected, a constant comparative method as proposed by Glaser and Strauss (1967) was adopted. We composed 'richly detailed' descriptions and accounts of the data collected, including field notes and the texts of the written forms (Denzin 1998, 335) to illustrate the participants' experiences in the programme. Categories and concepts were generated from each source of the data collected. Having processed the two data sources separately, the similarities and differences between the participants' experiences were identified and the relevant

categories and concepts generated from the two data sources were compared and contrasted. In short, the analytic procedure mentioned was a cycle of constant comparisons, moving in and out of the data, reading for meaning and advancing the analytical procedure through the intensive use of the above stages. Finally, similarities in the participants' experiences as generated from the data were categorised and used to construct this article.

Measures

Standard demographic characteristics including age, gender, meditation experience and religious faith were obtained. The participants from both the intervention and the control groups were invited to complete the following measures to assess the variables.

Mindfulness

Because we could not find appropriate mindfulness measures for young adults, we adapted the measures most commonly used for adults. Mindfulness was measured by the Mindful Attention Awareness Scale (MAAS) (Brown and Ryan 2003) and the Freiburg Mindfulness Inventory (FMI) (Walach et al. 2006).

The MAAS was chosen because it assesses the extent to which an individual is attentive to and aware of present-moment experiences during daily life. The scale is a 15-item self-report questionnaire with a 6-point Likert scale ranging from 1 (*almost always*) to 6 (*almost never*). Sample items include 'I find myself doing things without paying attention' and 'I find it difficult to stay focused on what's happening in the present.' A higher mean score reflects a higher level of mindfulness. The MAAS has a reported internal consistency of 0.87 (Brown and Ryan 2003). The Cronbach's alpha was 0.83, indicating high internal consistency.

The FMI 14-item form assesses the non-judgemental present-moment observations and openness to experiences of non-meditating samples. The 4-point Likert scale, which ranges from 1 (*rarely*) to 4 (*almost always*), measures three domains of mindfulness in daily life: mindful presence, openness to experience and non-judgemental acceptance. Sample items include 'I see my mistakes and difficulties without judging them' and 'I accept unpleasant experiences.' Higher scores are related to increased self-awareness and self-knowledge. The Cronbach's alpha was 0.753.

Well-being (WHO-5)

The WHO-5 (1998 version) was used for well-being assessment in this study (Bech 1998, 2001). It was derived from a large rating scale developed for a WHO project on quality of life in a group of diabetes patients (WHO-5 official website) and includes five items consisting of positive words such as positive mood, vitality and general interest. Each item is rated on a 6-point Likert scale from 0 (*at no time*) to 5 (*all of the time*). The total score range is from 0 to 25. Therefore, higher scores indicate better well-being. A score below 13 indicates poor well-being. Cronbach's alpha was 0.882, suggesting a good internal consistency.

Depression Anxiety Stress Scales (DASS)

The DASS is a 42-item instrument measuring the three dimensions of negative emotional states – depression, anxiety and stress – with 14 items for each domain (Lovibond and Lovibond 1993, 1995). A 4-point Likert-type scale ranging from 0 (*did not apply to me at all*) to 3 (*applied to me very much, or most of the time*) is used for the assessment. Sample items include 'I found it difficult to relax', 'I felt that life was worthwhile' and 'I felt I was close to panic.' A higher score indicates a greater level of distress. The internal consistency in the present sample for the depression, anxiety and stress scales was good (depression $\alpha = 0.903$, anxiety $\alpha = 0.776$ and stress $\alpha = 0.864$).

Perceived Stress Scale (PSS)

Perceived stress was measured by the Perceived Stress Scale (PSS) (Cohen, Kamark, and Mermelstern 1983; Cohen and Williamson 1988). The PSS is used to assess the degree of stress in one's life situation over the past month using 10 items. It is suggested that people examine potential challenging issues in relation to their available coping resources. A 5-point Likert-type scale ranging from 0 (*never*) to 4 (*very often*) is used. Sample items include 'How often have you felt that you were on top of things?' and 'How often have you felt nervous and "stressed"?'. A higher score indicates a greater degree of perceived stress. Cronbach's alpha was 0.545, which was relatively low.

All of the measures were translated into Chinese versions through a parallel back-translation procedure to achieve cross-cultural translation (Brislin 1986), except the WHO-5 and the DASS, which had already been translated and validated. With the exception of the PSS, all of the measures indicated acceptable internal consistencies. The low alpha score for the PSS was possibly due to difficulty understanding the translated items, particularly the negatively stated wordings. The participants' self-recorded journals were collected as qualitative data throughout the programme.

Results

Of the 70 final participants, 63 were female and 7 were male. Sixty-five participants were single, 4 were married and 1 did not specify any status. Twenty students were from Year 1 (67.1%), 6 were from Year 2 (8.6%), 13 were from Year 3 (18.6%) and 4 were from Year 4 (5.7%). Regarding religious affiliations, the majority were atheist (64.7%) and the rest were Buddhist (8.8%), Catholic (4.5%) and Protestant (22.4%). Two did not declare their religious affiliations. All of the participants were of Chinese ethnicity, and came from either Hong Kong or mainland China.

The baseline variable data were initially assessed using chi-square tests and independent sample *t*-tests. The chi-square results demonstrated that none of the variables differed significantly between groups: gender $\chi^2(1, N = 70) = 1.429$, $p = 0.428$; marital status $\chi^2(1, N = 70) = 2.138$, $p = 0.343$; subject major $\chi^2(6, N = 70) = 7.212$, $p = 0.302$; religious background $\chi^2(3, N = 70) = 0.585$, $p = 0.9$. The mean ages of the intervention and control groups were 21.74 and 22.89, respectively. Our findings from the *t*-test analysis indicate that there were no significant differences among the means of age and each variable between the intervention and control groups at the baseline ($p > 0.05$ in all cases). Table 1 presents the data from the *t*-test analysis.

Multivariate and univariate analysis of variance

The effects of the mindfulness programme were initially assessed using repeated-measures multivariate analysis of variance (MANOVA) to compare the intervention and control groups along the variables over time. The variables were Mindfulness, which was a combined measure of MAAS and FMI; Well-being, which was a measure of WHO-5; and Stress, which combined DASS and PSS. The two factors involved in the analysis were Group (intervention or control) and Time (baseline and post-intervention).

Mindfulness

The MANOVA showed a significant multivariate effect of Time on the combined measure Mindfulness (Pillai's F (2, 134) = 3.662, p = 0.028*, partial η2 = 0.052). However, there were no significant multivariate effects of Group (Pillai's F (2, 134) = 0.454, p = 0. 454, partial η2 = 0.636) and Group x Time (Pillai's F (2, 134) = 0.775, p = 0.463, partial η2 = 0.011). Each dependent variable was subjected to a further two-factor univariate ANOVA, the results of which revealed that there were no significant differences among the intervention and control groups over time on MAAS. However, after the programme, the MAAS score of the intervention group increased (from 3.89 to 3.97) while that of the control group dropped (from 4.07 to 3.93) in a coherent direction with the hypothesis. A significant difference in FMI score was found in the intervention group over time (Pillai's F (1, 136) = 5.297, p = 0.023*, η2 = 0.023). A post-hoc t-test revealed that the FMI scores of those who attended the mindfulness programme increased significantly (t(67) = –2.318, p = 0.024*), whereas the participants in the control group showed no significant change (t(68) = –0.913, p = 0.364).

Well-being

The WHO-5 scale scores at the baseline of both the intervention and control groups were below 13.0, which indicated poor well-being. The score increased at the post-test assessment. Two-factor univariate ANOVAs were performed to explore the effects of Time x Group on WHO-5. A significant difference in the effect on Time

Table 1. Independent sample t-tests on age and variables at baseline.

	Intervention (N = 35)		Control (N = 35)			
Variable	Mean	*SD*	Mean	*SD*	*t*	*p*
Age	21.74	3.06	22.89	6.32	–0.963	0.33
MAAS	3.89	0.54	4.07	0.56	–1.37	0.18
FMI	35.97	5.33	36.71	5.07	–0.60	0.55
WHO-5	12.54	4.65	12.49	4.2	0.06	0.96
DASS	28.31	14.16	26.91	11.83	1.4	0.66
PSS	25.20	2.98	24.4	3.15	0.8	0.28

Note: MAAS Mindful Attention Awareness Scale; FMI Freiburg Mindfulness Inventory; WHO Well-Being Index; DASS Depression Anxiety Stress Scales; PSS Perceived Stress Scale.
*p < 0.05; **p < 0.01; ***p < 0.001.

was found for WHO-5 (Pillai's F (1, 136) = 4.363, p = 0.039*, partial η2 = 0.031). The post-hoc t-test revealed that the WHO-5 scores of those who attended the mindfulness programme significantly increased (t(68) = –2.339, p = 0.022*), whereas the participants in the control group exhibited no significant change.

Depressive symptoms and perceived stress

No significant differences were found in the combined measurement of depressive symptoms and perceived stress between the intervention and control groups over time from the MANOVA. The two-factor univariate ANOVA also revealed no significant changes over time in DASS and PSS, respectively. However, the post-intervention means of both the intervention and control groups showed a consistent anticipated pattern in an expected direction for DASS and PSS. After the programme, the DASS score of the intervention group decreased from 28.31 to 22.53 while that of the control group increased from 26.91 to 27.74. The PSS score of both the intervention (from 25.2 to 24.79) and control (from 24.4 to 23.83) groups dropped. Therefore, the variables in general changed in a direction that was coherent with the hypothesis, but they did not reach significance.

Among the three dimensions of DASS for both the intervention and control groups, the mean score of depression (DASS-DEP), which was below 7, and the mean score of stress (DASS-STRESS), which was below 14, indicated a normal condition among the participants. However, the mean score of anxiety (DASS-ANX) was over 7 at baseline, which reflected that most of the participants experienced mild anxiety. The means and standard deviations of all of the variables at baseline and post-intervention for the intervention and control groups are presented in Table 2. Figure 1 plots the means of both the intervention and control groups, comparing pre- and post-test scores for each variable.

Correlation analyses

To examine the relationships between the variables, a series of correlations were calculated with the measures of mindfulness, well-being, depression, anxiety and stress before and after the intervention. Table 3 illustrates the correlation coefficients (values below the diagonal are for Time 1 and those above are for Time 2).

Table 2. Means and standard deviation at baseline and post-intervention for all variables.

	Intervention				Control			
	Pre		Post		Pre		Post	
Measures	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*
MASS	3.89	0.54	3.97	0.46	4.07	0.56	3.93	0.67
FMI	35.97	5.33	38.62	4.08	36.71	5.07	37.71	4.03
WHO-5	12.54	4.65	14.86	3.55	12.49	4.2	13.14	4.35
DASS	28.31	14.16	22.53	12.13	26.91	11.83	27.74	18.07
DASS-DEP	6.14	5.34	4.71	3.9	6.48	5.63	6.54	6.82
DASS-ANX	8.51	4.22	6.59	4.44	7.51	3.79	7.86	5.54
DASS-STRESS	13.66	6.49	11.4	4.91	12.91	5.58	13.34	7.39
PSS	25.2	2.98	24.79	2.57	24.4	3.15	23.83	3.02

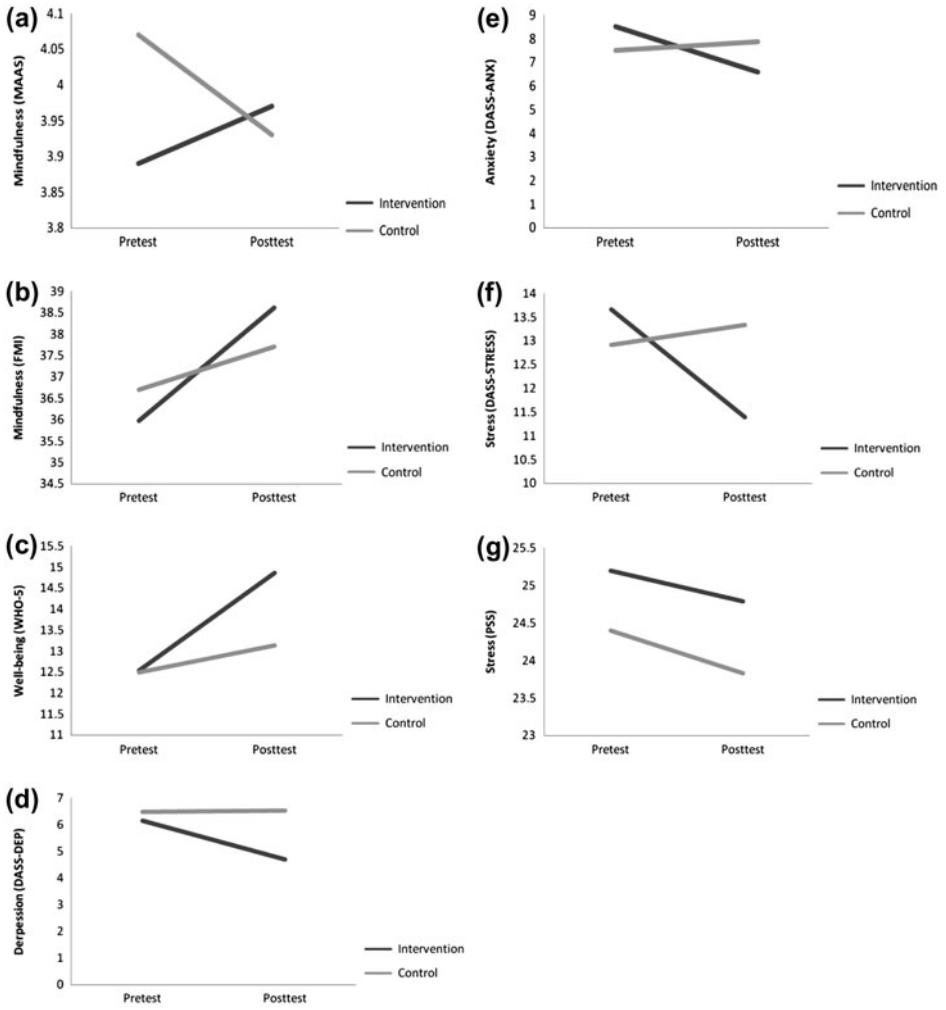

Figure 1. Plots of means displaying multivariate analysis by time interaction for both intervention and control groups. Graphs illustrate significant changes for the intervention: (b) Mindfulness (FMI), (c) Well-being (WHO-5). Graphs illustrate insignificant changes for the intervention: (a) Mindfulness (MAAS), (d) Depression (DASS-DEP), (e) Anxiety (DASS-ANX), (f) Stress (DASS-STRESS), (g) Stress (PSS).

Consistent with previous research, the general pattern of correlation indicates that mindfulness was positively correlated with well-being and negatively correlated with depression, anxiety and stress. Before the intervention, MAAS was only significantly correlated with DASS, DASS-ANX and DASS-STRESS. After the intervention, significant correlations were found between MAAS and all of the other variables. The coefficients were also larger after the intervention, which may indicate that the increase in mindfulness affected the variables. Similarly, FMI correlated significantly with all of the variables except PSS. Interestingly, PSS was not significantly correlated with any of the variables except MAAS. Moreover, among the

Table 3. Intercorrelations between mindfulness, well-being, depression, anxiety and stress.

		1	2	3	4	5	6	7	8
1	MAAS	–	.405**	.373**	–.559**	–.534**	–.515**	–.489**	–.286*
2	FMI	.387**	–	.420**	–.467**	–.450**	–.371**	–.399**	0.143
3	WHO-5	0.225	.532**	–	–.515**	–.471**	–.424**	–.505**	0.095
4	DASS	–.361**	–.524**	–544**	–	.885**	.919**	.937**	.269*
5	DASS-DEP	–0.219	–.466**	–479**	.820**	–	.709**	.720**	0.154
6	DASS-ANX	–.368**	–.313**	–.346**	.800**	.488**	–	.830**	0.231
7	DASS-STRESS	–.333**	–.499**	–.509**	.879**	.535**	.615**	–	.345**
8	PSS	–0.142	0.046	0.103	0.148	–0.06	0.153	.271*	–

Note: Values below the diagonal are for Time 1; values above the diagonal are for Time 2. ($N = 70$).
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

Table 4. Regression of mindfulness predicting well-being, depression, anxiety and stress at Time 1 and Time 2.

Predictor	Time 1	Time 2	Predictor	Time 1	Time 2
	Well-being (WHO-5)			Stress (DASS-STRESS)	
MAAS			MAAS		
B	1.79	2.64	*B*	–3.63	–5.41
SE	0.94	0.8	*SE*	1.25	1.17
β	0.225	0.37***	β	–0.33**	–0.49***
FMI			FMI		
B	0.452	0.42	*B*	–0.58	–0.62
SE	0.087	0.11	*SE*	0.12	0.16
β	0.532***	0.42***	β	–0.50***	–0.4***
	Depression (DASS-DEP)			Stress (PSS)	
MAAS			MAAS		
B	–2.16	–5.24	*B*	–0.79	–1.41
SE	1.17	1.00	*SE*	0.67	0.58
β	–0.22	–0.53***	β	–0.14	–0.29*
FMI			FMI		
B	–0.49	–0.62	*B*	0.027	0.1
SE	0.11	0.15	*SE*	0.072	0.09
β	–0.47***	–0.45***	β	0.046	0.14
	Anxiety (DASS-ANX)				
MAAS					
B	–2.68	–4.53			
SE	0.82	0.92			
β	–0.37**	–0.52***			
FMI					
B	–0.24	–0.47			
SE	0.09	0.146			
β	–0.31**	–0.37**			

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

intervention group, the number of hours one attended the programme was positively correlated with MAAS ($r = 0.260$; $p < 0.05$).

Regression analyses

Table 4 showed that the regression analysis of mindfulness was examined to reveal the ability of MAAS and FMI to predict well-being. In the regression with WHO-5 as the dependent variable, FMI was the only significant predictor at Time 1 (F (1, 68) = 26.83, $p < 0.001$). However, at Time 2, both MAAS (F (1, 68) = 10.98, $p < 0.01$) and FMI (F (1, 67) = 14.35, $p < 0.001$) were significant predictors.

The regression analysis of mindfulness was also examined to reveal the ability of MAAS and FMI to predict depression, anxiety and stress. In the regression with DASS-DEP, only FMI was a significant predictor at Time 1. At Time 2, both MAAS (F (1, 68) = 27.14, $p < 0.001$) and FMI (F (1, 67) = 17.06, $p < 0.001$) were significant predictors. In the regression with DASS-ANX, both MAAS and FMI were significant predictors at Times 1 and 2. However, the unstandardised regression coefficient of MAAS at Time 2 ($B = –4.53$, F (1, 67 = 24.22, $p < 0.001$) was higher than that at Time 1 ($B = –2.68$, F (1, 68) = 10.68, $p < 0.01$). The unstandardised regression coefficient of FMI at Time 2 ($B = –0.47$, F (1, 66) = 10.53, $p < 0.01$)

was increased compared with that at Time 1 ($B = –0.24$, F (1, 68) = 7.40, $p < 0.01$). In the regression with DASS-STRESS, both MAAS and FMI were significant predictors at Times 1 and 2. The unstandardised regression coefficient of MAAS at Time 1 ($B = –3.63$, F (1, 68) = 8.5, $p < 0.01$) was smaller than that at Time 2 ($B = –5.41$, F (1, 68) = 21.41, $p < 0.001$). However, in the regression with PSS, only MAAS was a significant predictor at Time 2 (F (1, 67) = 6.00, $p < 0.05$). Table 5 displays all of the statistics from the regressions of mindfulness on well-being, depression, anxiety and stress at Times 1 and 2.

A series of multiple regression models were constructed to examine the unique variance of the MAAS and FMI post-intervention. In each model, the MAAS was entered in the first step and the FMI was entered in the second step to check the unique variance in the outcomes beyond the variance accounted for by the MAAS. In the regression with WHO-5 as the dependent variable, the full model was significant ($R^2 = 0.12$, $F = 10.56$, $p < 0.01$). Entering FMI in the second step significantly improved the model ($R^2 = 0.2$, $F = 7.45$, $p < 0.01$) and, in the final model, both measures emerged as significant predictors.

In the regression with DASS-DEP as the dependent variable, the full model was significant ($R^2 = 0.27$, $F = 25.956$, $p < 0.001$). FMI significantly improved the model ($R^2 = 0.33$, $F = 6.76$, $p < 0.05$) and both measures emerged as significant predictors in the final model. In the regression with DASS-ANX as the dependent variable, the full model was significant ($R^2 = 0.25$, $F = 22.90$, $p < 0.001$). However, FMI did not improve the model significantly ($R^2 = 0.26$, $F = 2.44$, $p = 0.123$). In the regression with DASS-STRESS as the dependent variable, the full model was significant ($R^2 = 0.23$, $F = 20.95$, $p < 0.001$) and entering FMI in the second step significantly improved the model ($R^2 = 0.27$, $F = 4.5$, $p < 0.05$), such that in the final model both measures emerged as significant predictors. In the regression with PSS as the dependent variable, the full model was significant ($R^2 = 0.07$, $F = 5.7$, $p < 0.05$) and FMI significantly improved the model ($R^2 = 0.13$, $F = 6.3$, $p < 0.05$). Thus, both measures emerged as significant predictors in the final model. Table 5 shows all of the statistics for the multiple regression analyses of mindfulness predicting the variables post-intervention.

Participants' experience

Like the survey findings, the qualitative study consistently showed that all of the participants had a positive experience in the programme. The participants commonly reported that the programme's activities improved stress reduction, aroused a sense of compassion and helped them cultivate more awareness of the internal changes occurring in their minds and bodies. They also felt that the activities helped them identify the origins of their stress and gain some insight into how it might best be managed. The novel experiences provided by the programme clearly affected participants' mental and spiritual health in terms of enhancing their awareness of their emotional needs and nurturing their well-being and stress reduction. The effectiveness of the programme is reflected in the strong student demand that the programme continue. Participants commonly expressed missing the programme and confirmed that they would like to re-experience the benefits of the mindfulness activities. As one participant explained:

Table 5. Multiple regression analyses examining the contribution of mindfulness in predicting well-being, depression, anxiety and stress (Time = 2; $N = 70$).

Predictors		WHO-5		PSS	
		ΔR^2	Beta (β)	ΔR^2	Beta (β)
Step 1	MAAS	0.12**	0.37**	0.07*	–0.28*
Step 2		0.2**		0.13*	
MAAS			0.24*		–0.41**
FMI			0.32**		–0.31*

		DASS-DEP		DASS-ANX		DASS-STRESS	
		ΔR^2	Beta (β)	ΔR^2	Beta (β)	ΔR^2	Beta (β)
Step 1	MAAS	0.27***	–0.53***	0.25***	–0.51***	0.23***	–0.49***
Step 2		0.33*		0.26		0.27*	
MAAS			–0.41***		–0.43***		–0.39***
FMI			–0.28*		–0.18		–0.24*

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

> Frankly, Thursdays were my most exhausting day each week, [they were also] the most unsafe. However, after I attended the mindfulness session on these lowest mood days, I felt that I, as a whole person, was much more relaxed and I recovered my concentration and enthusiasm for study. I could also calmly deal with the negative emotions. Frankly speaking, as an exchange student, I came to a fully strange, big city. There were many difficulties and stresses, which I reluctantly faced – some difficult courses that I do not like… Indeed, to me, mindfulness is not only a stress-reduction programme. [The practice] has given me great insights into my value of life and personality. For example, [we were asked] to practise many new and strange stretching postures. The teacher asked us to accept the body sensations and feelings, and not to reject them. When we face various changing and impermanent situations, difficulties happen in our lives and we experience joy and sadness. These are just feelings, and we have to honestly accept them.

The psychological and spiritual aspects of the programme were beneficial, not only during the training sessions inside the Quiet Room, but also beyond. It was obvious that such effects could be extended into their daily lives by recalling memories of the activities involved in the programme and trying to transfer their benefits into everyday activities, such as having a slow lunch or sitting quietly in public transport vehicles. As the qualitative themes showed, the skills underlying the experience included slowing down on the pace of everyday life, a heightened sense of compassion, enjoying every 'here-and-now' moment and becoming aware of the internal changes in mind and body. All of these themes truly echoed the goals of the programme and identified its positive outcomes for the participants. The overall effectiveness of the programme can be seen in the following participant descriptions of the benefits of the various mindfulness activities:

> (1) Having a mindful lunch was the activity that I found most impressive. I felt relaxed and admired the food, which was made by my mother. We should care for every little detail of our lives. It is wonderful to eat lunch very slowly with others in a beautiful place.
> (2) Mindful listening was my favourite activity. When I was asked to be mindful about the sounds in the room, I was awakened to the sound of the bell. I learned how to observe the sound without judging it.
> (3) I am most impressed by the loving-kindness practice. It was pleasant. I felt warmth, love and hope in my heart. I also had the feeling of being loved. I am not used to expressing my love to my family and friends. Indeed, I love them, but without telling them.
> (4) Walking [was my favourite activity]. [It was] relaxed, comfortable and quiet. [I think that I] could walk more slowly in the daytime. [I realised] that I do not have to worry about what others think [about me], I just have to be myself. Mindful walking can enhance my abilility to observe tiny things carefully.

Discussion and conclusion

The study reported in this article was the first controlled trial of a mindfulness programme for pre-service teachers in Hong Kong. Our aim was to examine the possible effects of a six-week mindfulness programme for student teachers, and the feasibility of implementing the programme in a local community. Our preliminary findings provide positive evidence that the mindfulness programme improved the psychosocial situation. The study also suggests that a mindfulness-based programme is feasible for pre-service teachers.

The results indicate that most students experienced poor well-being and mild anxiety. However, the six-week mindfulness programme significantly increased the mindfulness and well-being of the intervention group. This suggests that mindfulness-based practices effectively enhanced the mental health and reduced the negative emotions of pre-service teachers. Our findings are consistent with previous studies in which there was a significant increase in mindfulness and well-being after an eight-week mindfulness programme (Baer 2003; Beitel, Ferrer, and Cecero 2005; Brown and Ryan 2003; Shapiro et al. 2008). Further, this study confirms the findings of previous studies on the effects of mindfulness activities on the promotion of some emotional and social skills, such as accepting experience without denying the facts and being calm and compassionate (Baer 2003; Salmon et al. 2004).

There were no significant differences among the DASS scores between the intervention and control groups over time. However, the depression, anxiety and stress scores of the intervention group dropped while those of the control group increased after the six-week mindfulness programme, suggesting that the changes may have been a result of mindfulness training. Moreover, the changes were coherent with recent prominent Western research that eight-week mindfulness training was a promising strategy for depression and anxiety reduction across clinical and nonclinical populations (Kabat-Zinn 1990, 1992; Kumar, Feldman, and Hayes 2008). In this study, insufficient mindfulness training in the six-week mindfulness-based programme may have caused the insignificant findings, but with reasonable changes in scores. A further study with a longer programme is necessary to gain a comprehensive understanding.

The correlation analyses revealed a positive correlation between mindfulness and well-being after the mindfulness programme, and a negative correlation between mindfulness and depression, anxiety and stress. However, a significant correlation between mindfulness (MAAS) and all of the variables was found only after the mindfulness programme. Most of the correlation coefficients between mindfulness and the variables increased post-test. The number of hours participants spent in the programme was significantly correlated with mindfulness (MAAS). In a previous study, an increase in mindfulness was found to mediate between formal mindfulness practice and well-being (Carmody and Baer 2008). Our outcome may indicate that the increase in mindfulness acted as a mediator in influencing the variables.

The results of the simple regression indicate that both MAAS and FMI were significant predictors of all variables. Likewise, there was significant correlation between the FMI and MAAS. In the multiple regression analyses, both MAAS and FMI accounted for significant and unique variance in predicting well-being, depression and stress. However, FMI did not act as a significant predictor of anxiety after MAAS was entered in the first block. This suggests that each measure assesses overlapping aspects of mindfulness, but with a unique facet of mindfulness that is not collectively captured. FMI in particular was designed for non-meditators. This pattern was similar to previous research (Shannon and Roemer 2012). In summary, an increase in mindfulness was predictive of an increase in well-being before and after the mindfulness training. A decrease in mindfulness predicted decreases in depression, anxiety and stress. Thus, addressing mindfulness can be a useful strategy for pre-service teachers to manage their depression and stress.

One interesting finding is that the correlations between PSS and the other variables were relatively lower. PSS was not correlated with all variables at baseline, and it was only significantly correlated with MASS negatively, and with

DASS-STRESS positively. The Cronbach's α coefficients of PSS were lower than expected, possibly owing to difficulties understanding the translated items, especially those that were negatively worded.

There were a number of limitations in this study. Small sample size was one of the major limitations. Although the students' recruitment response was unexpectedly enthusiastic, owing to scheduling difficulties there was a high drop-out rate in the beginning of the six-week programme. In the future, this problem can be solved by offering a formal course with credit bearing within the institution's schedule. Moreover, most of the students participating in this study were from Year 1, which did not include all of the typical psychosocial conditions of pre-service teachers. Student teachers generally have their most stressful experiences during their teaching practice at schools during their Year 3 and Year 4 studies. A study of Year 3 and Year 4 students engaged in teaching practice could effectively examine the psychosocial changes over that vital period. The lack of randomisation may also limit the study, as it is not possible to control the factor of motivation in practising mindfulness. We suggest conducting a randomised controlled study with a larger sample and long-term follow-up.

In conclusion, despite its limitations, this study has successfully served as the first attempt to examine the effects of a mindfulness programme for pre-service teachers in a Chinese community. This study provides initial and encouraging results for the potential effectiveness of mindfulness training in enhancing the psychosocial conditions of student teachers, with a significant increase in mindfulness and well-being. The positive responses from the qualitative findings reflected that a mindfulness-based programme was beneficial to student teachers and college students to increase well-being and reduce stress and anxiety. In our study, because mindfulness was a significant predictor of well-being, stress, anxiety and depression, a mindfulness programme can be considered a cost-effective stress management strategy for pre-service teachers during their professional training. Various school stakeholders, including teachers, school managers, policy makers and the teacher training providers, should be informed of the positive effects of mindfulness and its possible influence on school effectiveness and students' academic performance. Further, the abovementioned professionals should become more engaged in the professional dialogue on how the principle of mindfulness could be put into practice to enhance teachers' school experiences in an environment that is becoming increasingly stressful. At the level of teacher training, incorporating some elements of mindfulness training as basic outcomes of professional development programmes has great potential in courses promoting teachers' well-being and stress reduction. As this study identified, the three key domains of mindfulness to include are mindful presence, openness to experience and non-judgemental acceptance. The four skills of mindfulness are slowing down the pace of everyday life, cultivating a sense of compassion, enjoying every 'here-and-now' moment and becoming aware of the internal changes in mind and body. Overall, the inclusion of mindfulness training into teacher training programmes has the potential not only to enhance mental health, well-being and stress reduction, but also to reduce the potential risk of depression and burnout among teaching professionals. It is believed that this study could be replicated to provide further insight to be generalised across Western countries.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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